
Specialist Physical AI Need Not Be Opaque

A Fully Tensor-Decomposable VLA Policy

Protocol-aligned control and exact weight analysis

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Abstract

1 Tensor decomposition turns a large table of learned interactions into simpler low-
2 rank factors that can be ranked and inspected. Conventional robot policies contain
3 operations that block this direct weight-level view. We introduce χ -VLA, a 20.1M-
4 parameter vision-language-action policy built only from bilinear, linear, and rational
5 learned operations. Its weights can therefore be reorganized into explicit interaction
6 tensors rather than treated only as an opaque checkpoint. A mechanical audit and
7 numerical reconstructions verify that the deployed policy preserves this form.
8 Trained from scratch on LIBERO-Object, convolutional χ -VLA reaches $93.7\% \pm$
9 0.8% over three seeds under the published protocol. In a separate ViT instantiation,
10 we reconstruct every attention module in three checkpoints with worst relative
11 error 2.56×10^{-7} , and exact weight-derived subspaces beat equal-rank random
12 controls in all 36 primary block-head tests. The result is a capable and structurally
13 inspectable specialist family with exact access to trained attention coefficients. It
14 does not yet provide an exact end-to-end decomposition of the full policy.

15 1 Introduction

16 Modern robot policies can achieve high success while giving little indication of why an action was
17 chosen. Their checkpoints contain millions of weights, but those weights do not directly show which
18 image regions, words, and robot-state coordinates interact. Post-hoc probes can find correlated
19 activations, yet a correlation is not automatically a mechanism or a safe editing target.

20 A tensor is simply a multidimensional table. Tensor decomposition rewrites that table as a sum of
21 low-rank factors, analogous to finding dominant directions in a matrix. If every learned block uses
22 additions, multiplications, and ratios of polynomials, the full policy retains an explicit interaction
23 table defined by its weights. Its decomposed factors then provide candidate control mechanisms that
24 can be compared across layers and tested by intervention [3, 4]. This is what we mean by a fully
25 tensor-decomposable policy.

26 The restriction is nontrivial in robotics. Softmax, ordinary activations, and square-root normalization
27 break the algebraic form, while continuous actions require precise closed-loop outputs. The potential
28 payoff is also concrete: a weight-derived mechanism could help diagnose whether a policy follows
29 language, keys on a visual shortcut, or uses a fragile control feature.

30 We study the narrower but testable question:

31 *Can a specialist VLA retain strong closed-loop capability when every deployed*
 32 *learned operation is polynomial or rational, while supporting exact weight analy-*
 33 *sis?*

34 We test capability and structural analysis in two fully rational encoder instantiations. The convo-
 35 lutional checkpoints establish closed-loop capability. Separate ViT checkpoints support the exact
 36 attention audit.

37 Our contributions are:

- 38 1. We construct a complete VLA from tensor-compatible primitives. A rational normalization
 39 supplies input-specific scaling without leaving the rational function class.
- 40 2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with
 41 published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$
 42 over three seeds, and a three-checkpoint ensemble reaches 96.2%.
- 43 3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational
 44 and bilinear branches from their weights.
- 45 4. We extend exact weight-derived decomposition through individual attention layers with a
 46 reduced-Gram calculation that avoids materializing the full high-order tensor.

47 This paper does not claim current state of the art. It also does not claim that comparable mechanisms
 48 are impossible to obtain from conventional policies. The empirical question is whether the algebraic
 49 policy makes exact weight structure directly accessible.

50 2 A rational tensor robot policy

51 2.1 Tensor-compatible primitives

52 Each primitive replaces a familiar transformer component while keeping explicit coefficients. Lin-
 53 ear maps contribute first-order terms, while multiplying learned projections creates higher-order
 54 interactions that remain determined by the weights.

55 **Homogeneous coordinate.** We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative
 56 layer to represent bias, linear, and quadratic terms in one contraction.

57 **Bilinear feed-forward block.** For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

58 The inner sum is a table indexed by one output and two input coordinates. The three matrices give its
 59 CP factorization, the tensor analogue of a low-rank matrix factorization.

60 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top)/s] V. \quad (2)$$

61 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key,
 62 value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with
 63 interaction coefficients determined by the weights.

64 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
 65 root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z + \epsilon}$:

$$\text{RNorm}(x) = x r \left(\frac{1}{d} \sum_{j=1}^d x_j^2 \right). \quad (3)$$

66 Here rational means a ratio of polynomials. Projective coordinates carry each activation as a
 67 numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific
 68 rescaling without leaving the rational tensor representation.

2.2 Architecture and training

The policy follows the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven continuous action dimensions. The complete model has 20,137,352 learned parameters.

Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the trained checkpoint

Architecture definitions can differ from deployed code. We therefore audit the real 189/200 checkpoint at runtime and reconstruct representative operations from saved weights.

The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN reconstruction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed module’s float32 statistic. The audit certifies operator membership and local reconstruction, as summarized in Appendix Figure 3. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with:

- 50 canonical trials per task
- a 280-step cap
- three independently trained seeds per architecture
- 500 rollouts per seed and 1,500 rollouts per architecture

Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not rerun in our codebase, so differences may still include implementation and training choices.

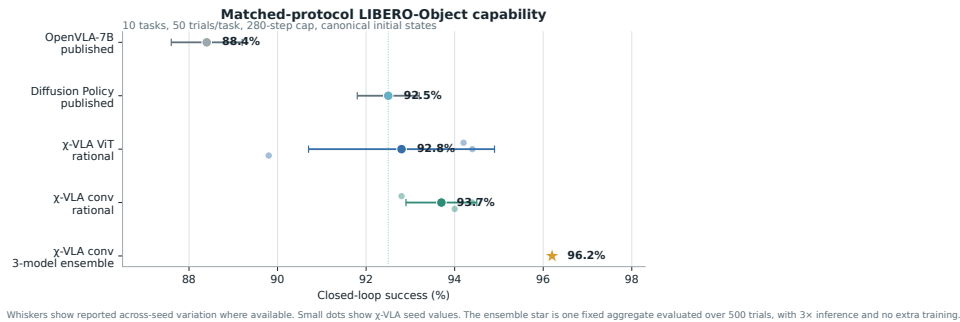


Figure 1: **Protocol-aligned LIBERO-Object capability.** Small dots are χ -VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

96 3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational conv	20.1M	3	93.7% $\pm$ 0.8%
χ -VLA, conv ensemble	$3 \times 20.1\text{M}$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	88.4% $\pm$ 0.8%
Diffusion Policy [1]	not matched	3	92.5% $\pm$ 0.7%

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

97 The convolutional seeds score 94.0%, 94.4%, and 92.8%. Their narrow spread shows that high
98 specialist capability is reproduced across three independently trained checkpoints.

99 Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3
100 points above the mean of its members and 1.2 points above the strongest member. The gain is largest
101 on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is
102 not one 20.1M-parameter policy.

103 3.3 What this result does and does not establish

104 The result shows that the rational architecture supports high closed-loop specialist capability. It is an
105 existence and reproducibility claim for a compact tensor-decomposable policy, not a claim of broad
106 robotic generalization or state of the art.

107 4 Exact weight-derived structure through attention

108 4.1 Exact layerwise construction

109 Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives
110 a coefficient table whose entries measure how combinations of query, key, and value coordinates
111 contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-
112 token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to
113 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

114 The deployed attention also normalizes each query and key with Equation 3. Each normalization
115 multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled
116 outside the bilinear dot products. The polynomial attention core is unchanged, while explicit
117 numerator and denominator factors carry the normalization. The rational extension agrees to $3.61 \times$
118 10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three
119 trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the
120 worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s
121 intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

122 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
123 values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors
124 rank important directions without constructing the full table. It matches brute-force materialization to
125 2.13×10^{-14} at toy scale.

126 4.2 Trained policy result

127 For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve
128 heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are
129 sampled once, then fixed and reused for all examples. The weights choose the subspace, while cached
130 activations are used only afterward to score how well it preserves the layer’s computation.

131 At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the
132 exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033
133 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks
134 0, 6, and 7.

135 This extends exact weight-only construction through individual attention layers. It is not an exact
136 end-to-end decomposition of the complete policy. Composition across all eight blocks still faces
137 rapid degree and representation growth.

138 5 Related work and limitations

139 **Related work.** OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control
140 baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative
141 weights can expose low-rank structure and support subsequent controls [3–5]. Recent work also
142 steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived
143 construction in a policy designed as a rational tensor program. Tensor and polynomial networks
144 provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard
145 LIBERO success can coexist with weak language use [9–11].

146 **Limitations and conclusion.** The benchmark is one specialist suite, so it does not establish broad
147 robotic generalization. Exact reconstruction is certified layer by layer and does not yet produce one
148 compact symbolic contraction of the full policy. The capability result uses convolutional checkpoints,
149 while the exact analysis uses separate ViT checkpoints. Evaluation uses fixed task instructions and
150 does not establish robust instruction grounding. Within that scope, rational restrictions support strong
151 LIBERO-Object control and exact trained-weight certificates. We do not claim a direct coefficient-edit
152 interface.

153 **Reproducibility.** The final artifact will provide code, configurations, checkpoint hashes, exact
154 certificates, and per-episode logs. Cache-backed operations join pinned LeRobot task metadata
155 before using official LIBERO task splits. Canonical-state checks fail loudly when requested states
156 are unavailable.

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182 A Supporting results and decisions

183 A.1 Per-task ensemble behavior

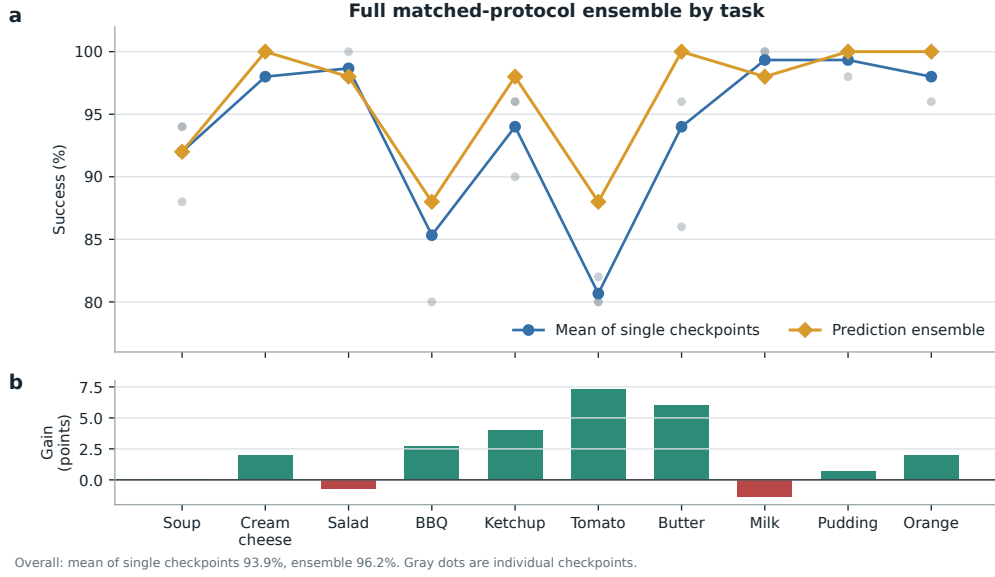


Figure 2: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

184 A.2 Expanded attention screen

185 The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128,
 186 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean
 187 is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96
 188 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider
 189 tested rank.

190 A.3 Numerical scope of rational conversion

191 The deployed Padé computation is exactly rational even though it approximates RMS normalization.
 192 Final certification must report its polynomial degrees, coefficients, denominator margin, observed
 193 activation range, tail error, action drift, and runtime overhead. Operator membership alone does not
 194 establish practical numerical safety under distribution shift.

Mechanical decomposability audit of the trained 189/200 checkpoint

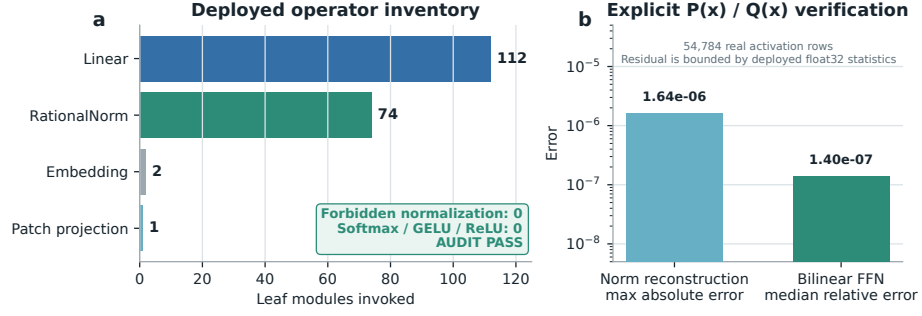


Figure 3: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.